Token Identity as a Routing Signal for Residual MLP Experts

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Abstract

Mixture-of-experts language models normally learn routing decisions from contextual hidden states. We study a narrower question: can token identity select a useful residual parameter subspace when a shared dense MLP remains active for every token? A fixed layer-specific lookup selects two narrow experts, while the shared branch and selected experts transform the same contextual hidden state. In a parameter- and token-matched single-seed comparison, 306.5M-parameter models are trained on 8B FineWeb-Edu tokens. The token-routed model reaches diagnostic-stream negative log-likelihood (NLL) 2.9329 versus 2.9482 for dense, but trains 21% more slowly. Independent-corpus evaluation changes ordering: routing is ahead on C4 validation and dense is ahead on a Pile test subset. Small Apple-MPS pilots additionally compare dense and routed feed-forward paths under both grouped-query and full multi-head attention. Their paired differences favor routing across the completed runs, but the pilots recycle a small local corpus and are not evidence of scaling or generalization. The results support token identity as a useful experimental routing signal, not as a general replacement for contextual routing.

1 Introduction

Sparse language models typically learn a router that maps each contextual hidden state to a subset of experts [2, 3, 5]. This combines two questions: whether conditional capacity is useful and whether a learned contextual decision is required to select it. Fixed routing separates these questions. Hash Layers showed that simple hashes of local token features can be competitive routing signals [4]; shared-expert systems suggest that common computation can remain dense while specialization is sparse [1].

We test a deliberately asymmetric design. Every token traverses a shared SwiGLU MLP. Token identity selects two additional narrow residual experts. Selection is context independent, but expert computation is not: both paths receive the contextual hidden state produced by attention. The study asks whether lexical identity can allocate a residual parameter subspace, not whether token IDs can replace contextual sequence processing.

Our main evidence is a matched 306.5M-parameter, 8B-token run pair. We report the complete scope of that observation, including slower routed training and opposite ordering on two independent corpora. We then add bounded local pilots that test whether the shared-plus-routed feed-forward design behaves similarly with grouped-query attention (GQA) and multi-head attention (MHA).

2 Architecture

For layer l, token ID t, and $E = 4$ stored experts, the primary experiment uses a seeded permutation of the token-ID residue,

$$r_{l,1}(t) = \pi_l(t \bmod E), \tag{1}$$

with cyclic secondary route

$$r_{l,2}(t) = \big(r_{l,1}(t) + 1\big) \bmod E. \tag{2}$$

The feed-forward output is

$$\begin{aligned}\mathrm{FFN}_l(\mathbf{x}, t) &= g_l^s\ \mathrm{Shared}_l(\mathbf{x}) \\ &\quad + g_l^r \sum_{j=1}^{2} 0.5\ \mathrm{Expert}_{r_{l,j}(t)}(\mathbf{x}),\end{aligned} \tag{3}$$

where shared and expert branches are SwiGLU transformations. The primary model learns scalar branch gates initialized to 0.5. The lookup has no learned router and receives no contextual state.

Let d_s be shared width, d_e expert width, E the number of stored experts, and k the number selected. Ignoring biases, stored intermediate width is $d_s + Ed_e$, while active width is $d_s + kd_e$. The primary routed model uses $(d_s, E, d_e, k) = (3840, 4, 64, 2)$, so its stored width matches the dense width of 4096 while its active width is 3968.

3 Primary 306.5M experiment

3.1 Protocol

The dense and routed models share an 18-layer decoder-only transformer, dimension 1024, 16 query heads, four

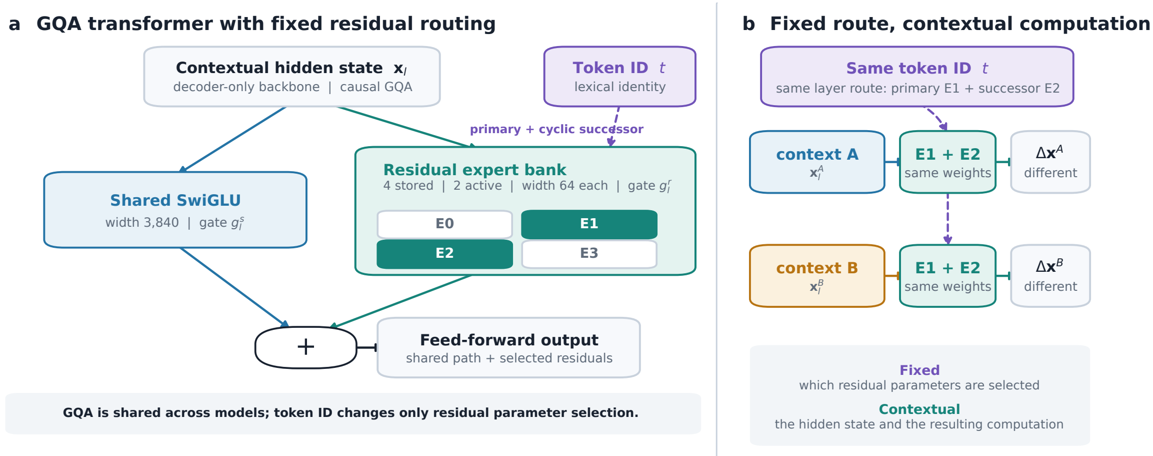

Figure 1: **Token identity controls parameter selection, not contextual computation.** Attention produces contextual hidden state $\mathbf{x}$. Every token traverses the shared SwiGLU branch. A fixed layer-specific table maps token identifier t to two narrow residual experts, which also transform $\mathbf{x}$.

Table 1: Primary matched comparison. Lower NLL is better.

Measurement	Dense	Token-routed
Eval-stream NLL	2.9482	**2.9329**
Final train NLL	2.9324	**2.9231**
Training throughput	~**0.95M**	~0.75M

Table 2: Independent-corpus NLL of the final checkpoints. Each row scores 262,144 targets in paired blocks.

Corpus	Dense	Routed	Δ
C4 validation	3.4161	**3.4066**	-0.0095
Pile test subset	**2.8690**	2.8769	$+0.0079$

key/value heads, QK-Norm, RoPE, a 32,000-token vocabulary, and context length 2048. They use the same FineWeb-Edu order, AdamW configuration, schedule, BF16 precision, and 8B-token budget. Each model contains 306.5M parameters. The comparison has one seed per architecture.

3.2 Matched-token result

At the last common diagnostic checkpoint, after 7.864B processed tokens, the routed model is ahead by 0.0153 NLL (Table 1). The diagnostic stream is fixed across models but drawn from the FineWeb-Edu training split; it is not held out. Routed training is approximately 21% slower in the reported implementation.

3.3 Independent-corpus evaluation

C4 favors routing, while the Pile subset favors dense (Table 2). No decontamination against FineWeb-Edu was performed. These measurements demonstrate corpus-dependent ordering rather than a universal generalization advantage.

4 Local Apple-MPS pilots

4.1 Purpose and protocol

The local pilots test the feed-forward design under two attention backbones. They are intentionally small architecture checks, not replications of the 8B-token experiment. Every run has 99,487,680 parameters, hidden size 384, 10 layers, eight query heads, and a 200,019-token o200k vocabulary. GQA uses two key/value heads; MHA uses eight. Each run trains for 250 steps with batch size 4 and sequence length 1024, totaling 1,024,000 sampled training tokens.

The source is a local 400-document FineWeb-Edu sample containing 356,120 tokens. A deterministic 95/5 split leaves 338,314 training tokens and 17,806 evaluation-tail tokens. Training therefore resamples the small training partition. Route frequencies are computed from the training partition only. Evaluation uses four batches sampled from the fixed tail, paired within each seed. All runs use AdamW with learning rate 3×10^{-4}, weight decay 0.1, and Apple MPS with PyTorch fallback kernels.

The local routed variant uses modulo-primary routing with a training-frequency-balanced secondary assignment, fixed top-2 weights 0.5/0.5, and no learned shared/routed gate. GQA allocates stored FFN width 1392+256 between shared and routed paths; its dense control uses width 1648. MHA allocates $1296 + 160$; its dense control uses width 1456. The differences preserve the same total parameter count within each paired comparison.

All four paired runs favor the routed feed-forward path (Table 3). The mean paired NLL difference is -0.049156 for GQA and -0.117579 for MHA. The substantial variation between MHA seeds cautions against interpreting

Table 3: **Matched local Apple-MPS pilots.** Δ is routed minus dense evaluation NLL within the same attention type and seed. Lower is better. The two seeds use the same split but different initialization and evaluation sampling seeds.

Attention	Seed	Dense NLL	Routed NLL	Dense PPL	Routed PPL	Δ NLL
GQA	42	7.596686	**7.536167**	1991.58	**1874.63**	−0.060519
GQA	43	7.530082	**7.492290**	1863.26	**1794.16**	−0.037792
MHA	42	7.586145	**7.536471**	1970.70	**1875.20**	−0.049674
MHA	43	7.726971	**7.541488**	2268.72	**1884.63**	−0.185483

either mean as a precise effect estimate. Current median logged MPS throughput is lower for routed models: 4708 versus 5418.5 tokens/s for GQA, and 4398.5 versus 5590.5 tokens/s for MHA at seed 42. These fallback measurements are implementation-specific and are not CUDA deployment benchmarks.

5 Limitations

The primary result has one seed per architecture. Its training-time diagnostic stream is not held out, and independent corpora change model ordering. The local pilots use two initialization seeds at most, recycle a small training partition, score only four sampled evaluation batches, and use an updated secondary assignment rather than the exact cyclic mapping of the primary model. Their widths were selected during local exploration. They therefore test implementation behavior and direction, not statistical significance, large-scale persistence, or downstream capability.

The strongest next tests are predeclared multi-seed training on a larger shard, frozen-checkpoint evaluation on separate corpora, equal-compute comparisons, and optimized sparse kernels. Learned contextual routers remain an important control: this work does not claim that fixed routing is generally superior to learned routing.

6 Conclusion

Token identity can provide a useful fixed signal for selecting narrow residual MLP experts while a shared dense path preserves common contextual computation. The 306.5M run pair shows a modest matched-token advantage but slower training and corpus-dependent generalization. Small local pilots show the same short-budget direction under both GQA and MHA, subject to much stronger limitations. The evidence motivates controlled replication rather than a claim of a universally better routing architecture.

Reproducibility. Code, configurations, model cards, and machine-readable metrics are available at https://github.com/Complexity-ML/complexity-framework. The primary checkpoints are available at https://huggingface.co/Pacific-i64/TR-MOE-306 and https://huggingface.co/Pacific-i64/Dense-306.

Use of generative AI tools. Generative AI tools assisted with language editing, consistency checks against author-provided metric files, figure scripting, and packaging. The author reviewed the numerical results, claims, source, and final text and takes responsibility for the work.

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